



An asymmetrical reinforcing spiral? Disentangling the longitudinal dynamics of media use and mainstream media trust

Yariv Tsfat¹, Rens Vliegenthart², Jesper Strömbäck^{3,*}, Elina Lindgren³

¹Department of Communication, University of Haifa, Haifa, Israel

²Strategic Communication Group, Wageningen University & Research, Wageningen, The Netherlands

³Department of Journalism, Media and Communication, University of Gothenburg, Gothenburg, Sweden

*Corresponding author: Jesper Strömbäck. Email: jesper.stromback@gu.se

Abstract

While numerous studies have documented an association between mainstream media trust and mainstream media use, only little is known about potential causal mechanisms underlying the association. We theorize that selective exposure, social influence, and the reinforcing spirals model offer three possible mechanisms that may underlie the association. These possibilities were studied using random intercept cross-lagged panel models and latent curve models on a four-wave panel data set ($n = 3,540$). For mainstream media, the results show an asymmetrical reinforcing spiral, in which the effect of mainstream media trust on use can be considered as one of yielding maintenance, while the impact of media use on mainstream media trust to a limited degree is in line with the notion of a spiral gradually (though inconsistently) increasing trust over time. We also find that use of right-wing alternative media consistently and sometimes robustly leads to decreasing mainstream media trust.

Keywords: reinforcing spirals model, news media trust, news media exposure, selective exposure, random intercepts cross-lagged panel modeling

The scientific exploration of audience trust in mainstream media has seen a substantial surge in recent years. Within this extensive body of research, a consistent finding is that audience trust in mainstream news media is positively correlated with mainstream news media exposure and negatively correlated with exposure to alternative, non-mainstream sources (Fletcher & Park, 2017; Jakob, 2010; Kalogeropoulos et al., 2019; Mourão et al., 2018; Prochazka & Schweiger, 2019; Tsfat & Cappella, 2003, 2005; Tsfat & Peri, 2006). These associations have been found in different countries with different media systems, and when using different measures of news media trust and exposure (Fawzi et al., 2021; Strömbäck et al., 2020).

However, only little is known about potential causal mechanisms underlying the association. In fact, in some studies, news media exposure is modeled as the dependent variable affected by news media trust (e.g., Fletcher & Park, 2017), while in others, news media trust is considered as the dependent variable affected by media exposure (e.g., Kalogeropoulos et al., 2019). The order of causality is motivated by different theoretical assumptions but lacks empirical testing. Hence, we do not know whether media use predicts media trust, whether media trust predicts media use, or whether the relationship can be characterized as mutually dependent, either in terms of positive reinforcing spirals (Slater, 2015; Slater et al., 2020; Thomas et al., 2021) or maintenance effects (Shehata et al., 2023).

This is problematic in itself, but also because it restricts knowledge-based measures towards increasing media trust. To be able to address public distrust in mainstream news media, we need to understand what causes it, and this includes knowledge of whether news media exposure is an outcome, or cause, of said trust.

Against this background, the purpose of this article is to investigate the longitudinal dynamics of media use and mainstream news media trust, taking the use of both mainstream news media and partisan, political alternative media into account. The longitudinal dynamics are studied using data from a four-wave, four-year panel survey inquiring into media use, news media trust, and related measures. Analytically, we apply random intercept cross-lagged panel models (RI-CLPM) and latent curve models. These approaches allow us to investigate the longitudinal dynamics at the individual level, which is particularly fitting when it comes to studying mutual relationships and reinforcing spirals (Thomas et al., 2021) and provides a stringent test of different types of mutual dependencies.

In brief, we find that the dynamics of the longitudinal relationship between media use and mainstream media trust are complex. In fact, for mainstream media an asymmetrical reinforcing spiral pattern emerges in which the effect of mainstream media trust on use can be considered as one of yielding maintenance, while the impact of media use on mainstream media trust to a limited degree is in line with the notion of a spiral gradually (though inconsistently) increasing trust over time. At the same time, we also find that use of right-wing but not left-wing political alternative media leads to decreasing trust in mainstream news media.

The complex relationship between media use and media trust

In general, trust is defined as the willingness of a trustor to be vulnerable to the actions of a trustee, irrespective of the ability to monitor or control that other party (Mayer et al.,

1995). The notion of mainstream media trust applies this definition to the context of the relationship between mainstream media and audiences. As mainstream media primarily cover the impersonal world, news media users lack the resources and abilities to thoroughly assess the reliability of news and journalists' interpretations of reality (Kohring & Matthes, 2007). In light of this uncertainty, the element of risk in defining trust remains pertinent in audience-media relations. Each decision made by audiences based on media information—whether it's voting, financial transactions, or altering travel plans due to media portrayals of health and security risks—can be seen through the lens of risk-taking (Fawzi et al., 2021; Strömbäck et al., 2020). If the news media breach this trust, it is likely that many of our day-to-day choices and behaviors could be viewed as wrong. Therefore, trust in mainstream media refers to the audience's perception that relying on mainstream media information is advantageous or at least not detrimental when making decisions. This is also reflected in how trust in mainstream media is conceptualized in recent review articles (Fawzi et al., 2021; Strömbäck et al., 2020).¹

As noted in the introduction, one consistent finding in research on the relationship between media use and mainstream media trust is that trust in mainstream news media is positively correlated with mainstream news media exposure and negatively correlated with exposure to alternative and non-mainstream sources² (Fletcher & Park, 2017; Jakob, 2010; Kalogeropoulos et al., 2019; Mourão et al., 2018; Prochazka & Schweiger, 2019; Strömbäck et al., 2020; Tsfati, 2010; Tsfati & Cappella, 2003, 2005). One key problem, however, is that most evidence is correlational and that only few studies utilize panel data that allows for inquiring into the longitudinal relationship between media use and mainstream media trust. While one implication is that our knowledge about the extent to which mainstream media trust affects media use, and media use affects mainstream media trust, is limited, another implication is that we only have limited understanding of the potential long-term dynamics.

From a theoretical perspective, at least three scenarios are possible. First, it might be the case that mainstream media trust is influencing media use, in the sense that those who are more trusting are more likely to use mainstream news media and less likely to use alternative or partisan media. Second, it might also be the case that media use is influencing mainstream media trust, in the sense that use of mainstream media leads to higher mainstream media trust, while use of alternative or partisan media leads to lower mainstream media trust. Third, building on the reinforcing spiral model (RSM) (Slater, 2015; Slater et al., 2020; Thomas et al., 2021), another possibility is that mainstream news media trust predicts media use, while also the reversed causal relationship is present. Below we will explicate the theoretical rationale for these scenarios.

Trust affecting exposure: Selective exposure

To begin with, the first theoretical possibility is that mainstream news media trust comes first and predicts subsequent media use. Along those lines, Tsfati and Cappella (2003) connect mainstream media trust and mainstream news exposure under the assumption that people are rational and aim to maximize their utility. They further posit that individuals turn to news sources in pursuit of accurate information about

the world. Acknowledging the notion of utility maximization and the impossibility of keeping up with all news channels constantly, they suggest that people have no choice but to be selective, and then choose those media that they find most trustworthy (Sears & Freedman, 1967; Stroud, 2011). Relaxing the assumption of rationality, trust may also function as a key heuristic when people choose what media to use.

Although the term “selective exposure” is most often applied to audience preferences for contents that support their political beliefs and attitudes, theoretically, the term is applicable whenever preferences influence the selection of particular information or information sources (Knobloch-Westerwick, 2014; Skovsgaard et al., 2016). For example, Wheelis (1974) found that when offered a list of public figures, people chose to be exposed to information from sources they trusted and rejected sources they mistrusted. Thus, the association between trust in mainstream media and the audience's news diet could be viewed through the prism of selective exposure theory, as an audience preference for information from trusted news sources. This theoretical perspective assumes that audiences scan the information environment and use their trust as a cue or heuristic that guides their media selections. If they trust mainstream news media, they use mainstream news media.

Selective exposure can also be used to explain the negative association between mainstream media trust and exposure to alternative, partisan sources. A negative correlation between partisan alternative media exposure and audience trust in mainstream media may reflect the tendency of mistrustful audiences to avoid or reject mainstream news media and instead seek alternative sources outside the mainstream (cf Andersen et al., 2023). While most research in this area is correlational, and thus does not provide evidence regarding the causal mechanisms, some empirical evidence supporting the “selective exposure” causal interpretation comes from qualitative studies on why people choose to consume partisan alternative media (Noppari et al., 2019; Schwarzenegger, 2023).

The logic of selective exposure implies that if we describe the association between mainstream media trust and media use over time, it is possible to hypothesize that (H1) trust in mainstream news media will positively predict later use of mainstream media (H1a) and negatively predict later use of partisan alternative media (H1b).

Exposure affecting trust: Establishing trust through repeated exposure, journalistic rhetoric and practices

As many have pointed out, a reverse causation in the association between trust in mainstream media and media exposure is also possible. Theoretically, a causal link from media exposure to mainstream media trust could be explained in terms of the establishment of trust through the participation in the ritual of repeated exposure to news (Silverstone, 2003). According to media sociologists, the embeddedness of television news in the everyday lives of the audience fosters a sense of ontological security that includes shared norms of trust (Allan, 2010, p. 110). The news media “invite us to trust them” (Silverstone, 1999, p. 123) and spending time with them may nurture and build a sense of identity as news consumers that includes trust in mainstream news media.

From a more psychological perspective, a causal link from media exposure to mainstream media trust could be explained in terms of exposure to trust-building journalistic

practices. Through their format, presentation, and content, news media work hard to establish the impression that they are trustworthy (Nelson & Kim, 2021; Robinson, 2023; Smeenk et al., 2023). Establishing the trust of the audience is the rationale behind many journalistic norms and practices from the rhetoric of objectivity and presenting conflicting views and relying on official sources (Tuchman, 1972) to presenting hard evidence and using transparency (Aharoni et al., 2023) and collaborative journalism strategies (Nelson & Kim, 2021). While many of these rhetorical devices have existed ever since the notion of objective journalism was developed in the 19th century, Smeenk et al. (2023) argue that more recent models of journalism, even those that employ more subjective writing, construct a journalistic ethos that aims at convincing audiences of journalists' trustworthiness (see also Robinson, 2023).

It therefore makes theoretical sense that the more audiences are exposed to such rhetorical devices, the more they may trust mainstream news. Experimental research has established that news reports are perceived as more trustworthy when they employ trust-building strategies (Borah, 2014; Henke et al., 2020). Such experimental research "focuses on the micro level and largely relies on individual characteristics and perceived quality of the communicator as well as the news item itself, based on multiple factors like fairness, objectivity, accuracy, and believability" (Henke et al., 2020, p. 301). Although evidence is mixed in terms of how transparency influences people's media trust (Karlsson, 2021), repetitive exposure to such journalistic devices may generalize to macro-level trust that concerns the relationship between the audience and the media as an institution. In other words, repeated exposure to journalistic practices and rhetorical devices may affect attitudes and cultivate trust through influencing audiences' expectations of what constitutes proper news journalism in terms of the format, tone, and editing.

The observed *negative* association between alternative media exposure and trust in mainstream media may instead be explained by more straightforward persuasion. Criticisms and attacks on mainstream news media are seen by some as a defining characteristic of many partisan alternative media (Tsfati & Peri, 2006, p. 169). By investing efforts in attacking, delegitimizing and criticizing mainstream media sources (Chadha & Bhat, 2022; Jamieson & Cappella, 2008), partisan media try to shape audience attitudes toward mainstream media and provide audiences with reasons as to why mainstream media should not be trusted. Previous research has also found that attacks on and criticisms of mainstream media are prevalent in partisan alternative media (Cushion, 2024; Figenschou & Ihlebaek, 2021; Jamieson & Cappella, 2008; Roberts & Wahl-Jorgensen, 2020)—also in the case of Sweden (Ljunggren Forsberg, 2021). Given research documenting that exposure to anti-media criticism reduces audience trust in media (Guess et al., 2021; Watts et al., 1999), it makes sense to expect that use of partisan alternative media may decrease audience trust in mainstream media.

These theoretical perspectives imply that if we describe the association between media use patterns and mainstream media trust over time, it is possible to hypothesize that (H2) earlier media use predicts later trust in mainstream media. More specifically, earlier use of mainstream media can be expected to increase mainstream media trust (H2a), while earlier use of partisan media can be expected to decrease trust in mainstream media (H2b).

Reciprocal relationship between trust and exposure: The reinforcing spirals model

Third, the association between trust in mainstream news media and media exposure patterns may be dynamic and reciprocal across time, that is, mainstream media trust may shape future media exposure and at the same time be shaped by it. This possibility is consistent with Slater's (2007, 2015) Reinforcing Spirals Model (RSM), which has become one of the most influential models of media use and media effects in contemporary research (Slater et al., 2020). The RSM utilizes a systems-theory perspective "for conceptualizing media use as part of a dynamic, endogenous process combining selective exposure and media effects" (Slater, 2015, p. 370). According to the RSM, prior attitudes lead to exposure to media content consistent with these attitudes (*selection effect*). This exposure, in turn, may confirm and strengthen the prior attitudes (*media effect*), thereby making them stronger (that is, more polarized and extreme). The updated attitudes again result with exposure to attitude-congruent media messages, which may further polarize opinions until an equilibrium is reached.

Applying this model to the association between mainstream media trust and media use implies a reciprocal relationship in which mainstream media trust shapes future media use, while at the same time current media use simultaneously affects future trust in mainstream media.

Important to note though is that such a spiraling process can take two forms (Slater, 2015). Normally, Slater (2015) argues that the RSM predicts an "homeostatic process or a negative feedback loop" (p. 373) in which the mutual effects of exposure on attitudes and vice versa results in what is also known as a maintenance effect (Shehata et al., 2021; Shehata et al., 2023). Such a process means that a feedback loop enhances a situation where media use *prevents* a change process that would occur in the absence of exposure to certain media (Shehata et al., 2023, p. 3). If mainstream media trust and media use are longitudinally related but without changes in levels of media use or mainstream media trust across time, it would thus suggest such a homeostatic type of reinforcing spiral and maintenance effect. The less regular situation predicted by the RSM, according to Slater, is characterized by "continuing and even nonlinear growth over time in a given type of media use and outcome" (Slater, 2015, pp. 272–273). That may lead to a positive feedback loop in which both exposure and attitudes are gradually driven toward more extreme levels. In the case of media trust and use over time, this would mean increases in both trust in and use of mainstream news or, in the case of partisan media, increasing use of those accompanied by decreasing trust in mainstream news media.

Spiraling effects that "grow out of control" can, however, be expected to be rather uncommon. According to Slater (2015, p. 373), such effects are "most likely only for those particularly at risk due to social circumstances or personality-based vulnerabilities." Even so, it is a theoretical possibility. While the RSM model was introduced more than 15 years ago (Slater, 2007), some of the results point out to rather complex dynamics (Geusens & Beullens, 2021) and the pattern is almost never a perfect spiral (e.g., Dahlgren et al., 2019; Kruikemeier & Shehata, 2017; Slater et al., 2020; Strömbäck & Shehata, 2019).

To explore the long-term dynamics in the relationship between media use and mainstream media trust, we therefore ask (RQ1): Does the longitudinal relationship between

mainstream media trust and media use represent a case of a positive feedback loop in which audiences are driven to extreme levels of media trust and use, or a case of maintenance effects in which the hypothesized mutual relationship results in homeostasis?

Case selection, data, and methods

To investigate our hypotheses and address our research question, we focus on the case of Sweden. This case was chosen for several reasons. First, mainstream news media are both widely used and rather trusted in Sweden (Martinsson & Andersson, 2022; Tsfati et al., 2022, 2023). Second, over the last decade, partisan political alternative media have become more prominent and widely used (Heft et al., 2020; Holt, 2020; Newman et al., 2023). With respect to right-wing political alternative media, research also suggests that Sweden has a more established infrastructure and user base than in many other countries (Heft et al., 2021). This combination is important since it suggests that those who have lower trust in mainstream media have the option to seek out political alternative media both to the left and to the right. Thereby, this case provides the necessary condition for investigating the longitudinal relationship between mainstream media trust and use of both mainstream news media and partisan political alternative media. Third, research shows that in Sweden, there is a clear distinction between mainstream, traditional news media on the one hand and partisan, political alternative media on the other, and that when citizens are asked about “news media,” they are thinking about traditional, mainstream news media (Tsfati et al., 2023).

Also important is that selecting Sweden allows us to utilize a unique four-wave panel study carried out between 2020 and 2023, thereby enabling us to investigate the longitudinal relationship between mainstream media trust and media use.

Data

The four-wave panel study used here was designed by the authors. The first wave of the survey was administered between February 24 and March 24, 2020 (98% of respondents completed the survey before the Swedish Public Health Agency responded to the first indications of community transmission of the Corona virus by advising social distancing for the elderly on March 16, 2020). The subsequent waves were administered respectively in 2021 (February 25–March 28), 2022 (February 24–March 31) and 2023 (February 28–March 23).³ The panel survey was carried out by the SOM institute, a Swedish survey research infrastructure at the University of Gothenburg. The survey comprises a probability-recruited sample and a non-probability sample, and pre-stratification was applied to ensure that the invited samples resembled the Swedish population in terms of the key demographics age, education, and gender. In the present study, we include the respondents from both sample types who completed all four waves of the panel study ($n = 3,540$). Details about data collection, response, and attrition rates are provided in the [Supplementary Appendix A](#) and about the demographic breakdown in [Supplementary Appendix B](#).

Measures

News media trust was measured using an abbreviated version of Gaziano & McGrath’s (1986) News Credibility Scale. As suggested by Strömbäck et al. (2020), this is probably

the most popular and efficient trust in media scale, which has been extensively validated in previous research (e.g., Tsfati, 2003; Tsfati & Peri, 2006; West, 1994). Question wording asked whether the Swedish news media are “fair,” “unbiased,” “tell the whole story,” “accurate,” and “separate fact and opinion” in their news coverage. The response categories ranged between 1 = “strongly disagree” and 7 = “strongly agree.” The items loaded on a single factor in all waves. Factor analyses are presented in the [Supplemental Appendix C](#). The scale was calculated as the mean of the five items ($M_{\text{wave1}} = 3.99$, $SD = 1.44$, Cronbach’s $\alpha = .94$; $M_{\text{wave2}} = 3.93$, $SD = 1.39$, $\alpha = .93$; $M_{\text{wave3}} = 4.16$, $SD = 1.40$, $\alpha = .93$; $M_{\text{wave4}} = 4.00$, $SD = 1.40$, $\alpha = .94$). For robustness checks, we also used a single-item measure of trust in the Swedish media, worded “Generally speaking, to what extent do you trust information from the news media in Sweden?” Answer categories varied between 1 for “do not trust at all” and 7 for “trust completely” ($M_{\text{wave1}} = 4.85$, $SD = 1.30$; $M_{\text{wave2}} = 4.89$, $SD = 1.30$; $M_{\text{wave3}} = 4.94$, $SD = 1.29$; $M_{\text{wave4}} = 4.77$, $SD = 1.88$). The robustness checks using this single item media trust measure are reported in the [supplementary material](#), as Model 2 in the three tables composing [Supplementary Appendix H](#).

As mentioned above, important to note is that previous research has found that when people are asked about their news media trust, they are thinking about traditional, mainstream news media (Ladd, 2011; Tsfati & Peri, 2006). This was also the case in Sweden in particular, both for a single-item media trust measure and for a media trust scale (Tsfati et al., 2023). To test this assumption empirically, in Wave 2 of the survey, respondents were asked specifically and separately about their trust in each of the outlets in our exposure scales. Both measures of trust in media were positively and strongly correlated with a measure of trust in the specific mainstream outlets in our exposure scale ($r = .78$ for the single item, and $r = .72$ for the News Credibility Scale). The association between trust in the left-wing outlets and both measures of mainstream news trust was much smaller ($r = .38$ for the single-item measure; $r = .41$ for the scale), and even negative for trust in right-wing outlets ($r = -.17$ for the single item measure; $r = .19$ for the scale; $p < .001$ for all these correlations).

Mainstream news media exposure. We measured individuals’ exposure to both mainstream and partisan alternative media by asking the following question: “In a typical week, how often do you use the following news media, in their traditional formats or online?” (with answers coded 1–8, with 1 meaning “never/more seldom” and 2–8 indicating 1–7 days a week) followed by a list of news outlets. The list of outlets and information about them can be found in the [Supplementary Appendix D](#). Exploratory Factor Analysis of the eight mainstream outlets (reported in the [Supplementary Appendix E](#)) shows that the TV outlets loaded on one factor, the tabloids on a second, and the daily newspapers together with the public radio station on a third. As all eight outlets have been found to be understood by the public as mainstream outlets (Tsfati et al., 2023), and for the sake of simplicity, we calculated a single measure of exposure to mainstream outlets (*Aftonbladet*, *Expressen*, *Dagens Nyheter*, *Svenska Dagbladet*, *Rapport in SVT*, *Aktuellt in SVT*, *Nyheterna in TV4*, and *Ekonyheterna in Sveriges radio*). Separating the exposure measures in line with the factor analysis results yielded similar patterns to the ones reported

below. These results are reported as robustness checks in the [Supplementary Appendix H](#). Exposure to mainstream outlets was calculated as the mean of the eight items and then scaled to a 1–7 range, to match the scale of the media trust measures ($M_{\text{wave1}} = 3.27$, $SD = 1.18$, $\alpha = .54$; $M_{\text{wave2}} = 3.35$, $SD = 1.31$, $\alpha = .55$; $M_{\text{wave3}} = 3.46$, $SD = 1.33$, $\alpha = .55$; $M_{\text{wave4}} = 3.33$, $SD = 1.35$, $\alpha = .58$).⁴

Partisan alternative news media exposure. The same exposure question was asked about a list of non-legacy, online alternative media outlets and economic news outlets. Factor analyses of exposure to all those 15 alternative outlets yielded incoherent and inconsistent results. The only consistent findings were that right-wing (*Fria Tider*, *Sambällsnytt* and *Nyheter idag*) and left-wing political alternative media (*Arbetaren*, *Arbetet*, *ETC* and *Expo*) loaded separately and consistently across waves. An EFA of these 7 items (reported in the [Supplementary Appendix G](#)) yielded consistent factors of exposure to right-wing and left-wing outlets. Therefore, we calculated measures of exposure to right-wing ($M_{\text{wave1}} = 1.36$, $SD = 1.15$, $\alpha = .90$; $M_{\text{wave2}} = 1.27$, $SD = .95$, $\alpha = .86$; $M_{\text{wave3}} = 1.24$, $SD = .95$, $\alpha = .82$; $M_{\text{wave4}} = 1.21$, $SD = .81$, $\alpha = .66$) and left wing ($M_{\text{wave1}} = 1.25$, $SD = .80$, $\alpha = .78$; $M_{\text{wave2}} = 1.15$, $SD = .55$, $\alpha = .66$; $M_{\text{wave3}} = 1.14$, $SD = .51$, $\alpha = .65$; $M_{\text{wave4}} = 1.14$, $SD = .49$, $\alpha = .66$) outlets. We again calculated the mean of the items and then scaled the resulting scores to a 1–7 scale, to match the news media trust measures.

Analysis

To investigate the relationship between media use and trust, we rely on random intercept cross-lagged panel models. This approach is considered particularly fitting when investigating mutual relationships and reinforcing spirals ([Thomas et al., 2021](#)). In panel analyses, a distinction between inter (*between*)—and intra (*within*)—individual effects can be made. The first can be attributed to structural differences across individuals in media use and mainstream media trust. The second captures over-time variation *within* individuals and thus the over-time dynamics that underlie the idea of reinforcing spirals. In regular cross-lagged panel models, inter-individual differences are imperfectly captured through the lagged dependent variables and potentially the inclusion of control variables. Consequently, it is impossible to fully distinguish between inter- and intra-individual effects.

Therefore, we include random intercepts for both variables of interest to capture the variance across individuals, and with the models we focus on intra-individual variance and thus the over-time dynamics between exposure and trust. Reinforcing spirals are present in instances where we find mutual dependent relationships that go in the same direction. A more stringent test of the spiral notion that is based on positive feedback loops would additionally require a growth trajectory for both variables. If such a growth is not present, the relationship between the two variables should rather be described as one of homeostasis or maintenance ([Shehata et al., 2023](#)). We test whether we find such a growth trajectory by testing whether a linear slope is present in models that predict mainstream media trust and media use. Initially, we employ a constrained random intercept model for each outcome individually, which assumes over time stability. In a next step, we augmented both models by introducing a random linear slope, leading to a latent curve model (LCM). The LCM provides

insights into the over-time variation and models the average linear growth rate (see [Curran et al., 2014](#)). In instances where the slope significantly contributes to a better model fit, we include a linear slope in the final models as presented below.

Ultimately, we run separate analyses for mainstream media use, and alternative right-wing and left-wing media use, using the *Lavaan* package in R. For each, we run three models. First, we ran a regular cross-lagged panel model (CLPM). Second, we ran a RI-CLPM in which random effects are captured by a latent variable accounting for individual differences ([Hamaker et al., 2015](#)), and where the analyses thus focus on intra-individual dynamics. Third, we ran another RI-CLPM where we constrained effects between wave one and two, and two and three, respectively, to be equal to test stability over time. We compare model fit statistics and present the results from the best fitting model. If needed, we add a linear slope to the selected model. Finally, to test whether effects of media use on mainstream media trust are stronger than of mainstream media trust on media use, or vice versa, we ran a final model where we constrained coefficients to be similar for both directions. We only do that where a positive and significant effect is present in at least in one of the cases.

Results

To test our hypotheses, we focus on the various variants of the cross-lagged panel model. We first test which model is most appropriate. For each type of media use, we compared cross-lagged panel models (CLPM) with random intercept cross-lagged and the option to constrain effects to be similar over-time (RI-CLPM). In all instances, the RI-CLPM demonstrated a better model fit than the regular CLPM (for mainstream media use $\Delta \chi^2 = 2189.6$, $df = 3$, $p = .000$, for alternative right-wing media use $\Delta \chi^2 = 1543.6$, $df = 3$, $p = .000$, and for alternative left-wing media use $\Delta \chi^2 = 2055.8$, $df = 3$, $p = .000$). In the RI-CLPM, between-person effects are captured by a latent variable that represents a person specific random intercept. In all instances, the results show that the regular RI-CLPM outperforms the one that constrains effects to be similar over-time (for mainstream media use $\Delta \chi^2 = 247.0$, $df = 14$, $p = .000$, for alternative right-wing media use $\Delta \chi^2 = 412.5$, $df = 14$, $p = .000$, and for alternative left-wing media use $\Delta \chi^2 = 366.5$, $df = 14$, $p = .000$).

Results from analyses where a linear slope is added to a constrained random intercept model for each outcome individually indicate that a growth trajectory is present for mainstream media trust ($b = .019$, $se = .006$, $p = .001$, with significant variation across individuals: variance = .053, $se = .004$, $p = .000$), but not for media use ($b = -.011$, $se = .019$, ns). We consequently add a linear slope for mainstream news media trust, but not for media use, to the final models.

The findings from the final RI-CLPM for mainstream media are presented in [Figure 1](#) (with full details reported in the [Supplementary material](#) as Model 1 in [Supplementary Appendix Table H1](#)). Model fit statistics are satisfactory. While the Chi-squared is significant (Model 1: $\chi^2 = 113.730$; $df = 8$, $p = .000$), the other fit statistics indicate a satisfactory model fit (For Model 1: CFI = .995; RMSEA = .029). The results are largely in line with H1a. We find that in all three instances (wave 1 to wave 2: $b = .098$, $se = .028$, $p = .000$; wave 2 to wave 3: $b = .087$, $SE = .042$, $p = .040$; and wave 3 to wave 4: $b = .081$, $SE = .030$, $p = .007$),

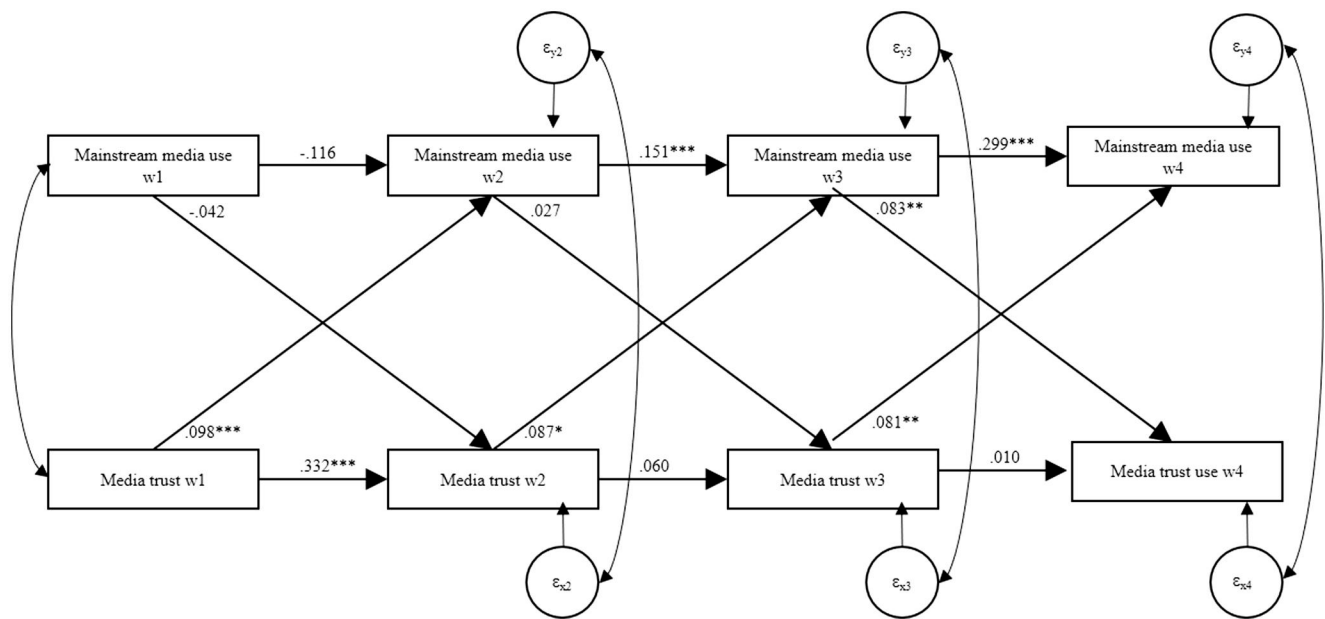


Figure 1. Reciprocal relationships between mainstream media trust and mainstream media use.

Note. RI-CLPM with a linear slope for media trust, unstandardized B values, $\chi^2=113.730$; $df=8$, $p<.001$, CFI=.995.061; RMSEA=.06129; * $p<.05$; ** $p<.01$; *** $p<.001$, $N=3,540$.

mainstream media trust yields higher levels of subsequent mainstream media use (H1a). For the reversed relation, we have findings that are partly in line with the hypothesis, although the only significant coefficient is for wave 3 use on wave 4 trust (wave 3 to wave 4: $b=.083$, $SE=.030$, $p=.005$). This partially confirms H2a.

The coefficients indicate that the impact of media use on mainstream media trust from wave 3 to 4 is comparable to the reversed effect. A model that constrains these effects to be similar indeed does not lead to a worse model fit ($\Delta\chi^2=.801$, ns). However, overall, mainstream media trust has a more consistent effect on media use than the other way around.

Our RQ asked whether the relationship between mainstream media use and trust can be considered as one of a reinforcing spiral in which audiences are driven to higher levels of trust and use or whether it resembles a maintenance effect. The results above indicate that neither is completely the case. More specifically, the impact of mainstream media trust on use can be considered as one of yielding maintenance (as no linear slope is present), while the impact of mainstream media use on mainstream media trust to a limited degree is in line with the notion of a spiral gradually increasing trust over time (a significant linear slope, but only some of the effects are significant). The reasons for these patterns, which we call an asymmetrical reinforcing spiral, and its theoretical ramifications will be discussed in the discussion section.

We now move on to the hypotheses targeting the interactions between mainstream media trust and partisan alternative media use (both right-wing and left-wing). Here, we anticipated negative reciprocal effects and thus a negative reinforcing spiral. We already know that mainstream media trust demonstrates a growth trajectory, and we capture that by including a linear slope in the models. For right-wing media use, we find that such an over-time trajectory is present ($b=-.032$, $se=.010$, $p=.002$, with significant variation across individuals: variance = .070, $se=.033$, $p=.035$). The negative direction indicates a downward over-time trajectory.

Thus, we include (co-varying) linear growth slopes for both mainstream media trust and right-wing media use. For left-wing media, we do not find a significant linear slope ($b=-.002$, $se=.004$, ns). Figures 2 and 3 present the results of the final models (full details are again presented in the [Supplementary material](#), as Model 1 in the tables composing [Supplementary Appendix H](#)).

The results partially corroborate our expectations. When it comes to the impact of mainstream media trust on right-wing media use, we find some support for H1b, although only the effect of wave 3 trust on wave 4 use is significant ($b=-.109$, $SE=.046$, $p=.018$). Results for the impact of right-wing alternative media use on trust in mainstream media align more with our expectations. More specifically, we find that right-wing media use consistently decreases trust in mainstream media in all three cases (wave 1–wave 2: $b=-.078$, $SE=.020$, $p=.000$; wave 2–wave 3: $b=-.135$, $SE=.049$, $p=.005$; wave 3–wave 4: $b=-.370$, $SE=.155$, $p=.017$). This is in line with H2b. For left-wing media, we do not find any significant associations between media use and mainstream media trust. Thus, we can partially confirm H2b.

Robustness tests

To test for the robustness of these results, we ran the same models with the single-item trust measure instead of the news credibility scales. The resulting patterns (reported in [Supplementary Appendix Tables H1–H3](#)) were altogether similar, in terms of the significance tests, coefficient sizes, and growth curves and model fit.⁵ Here, we also report results that look specifically at TV news use and tabloid use ([Tables H4 and H5](#)). While growth curves and model fit demonstrated identical patterns for both TV news use and tabloids, significance tests for the coefficients in these analyses yielded somewhat weaker results when examining these different types of mainstream news media separately. These weaker associations can be explained by the fact that our trust

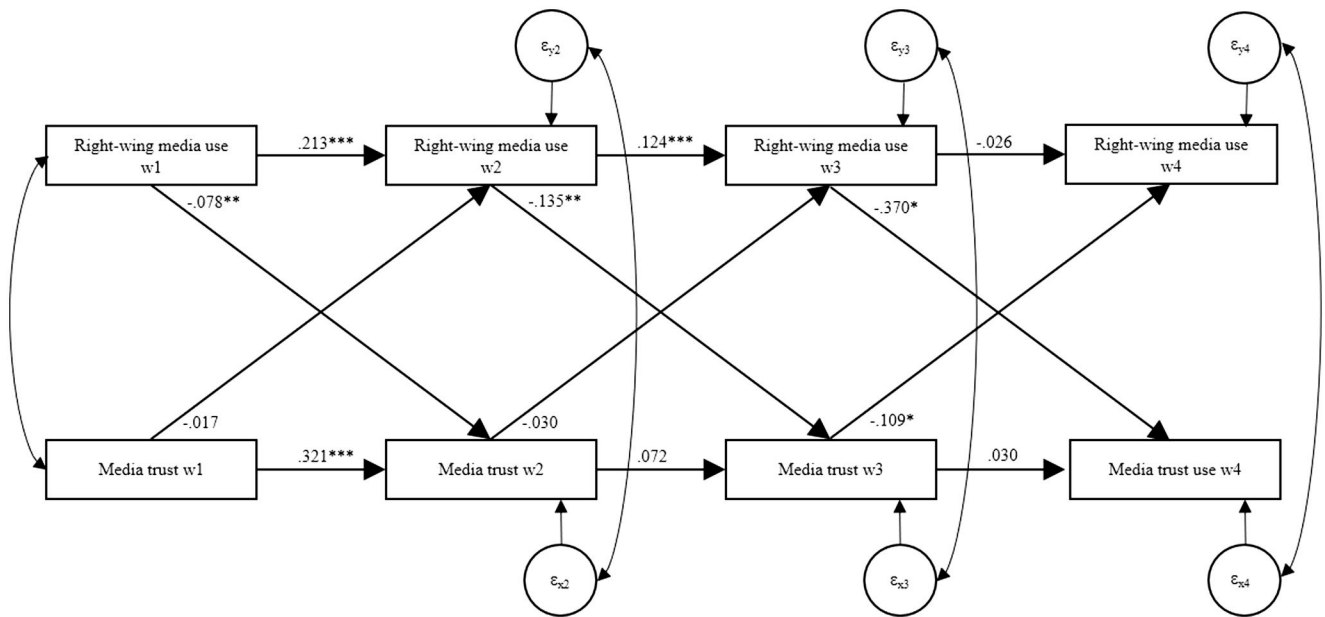


Figure 2. Reciprocal relationships between mainstream media trust and alternative right-wing media use.

Note. RI-CLPM with linear co-varying slopes for media trust and use, unstandardized B values, $\chi^2=101.688$; $df=6$, $p<.001$, CFI=.996; RMSEA=.067; * $p<.05$; ** $p<.01$; *** $p<.001$, $N=3,540$.

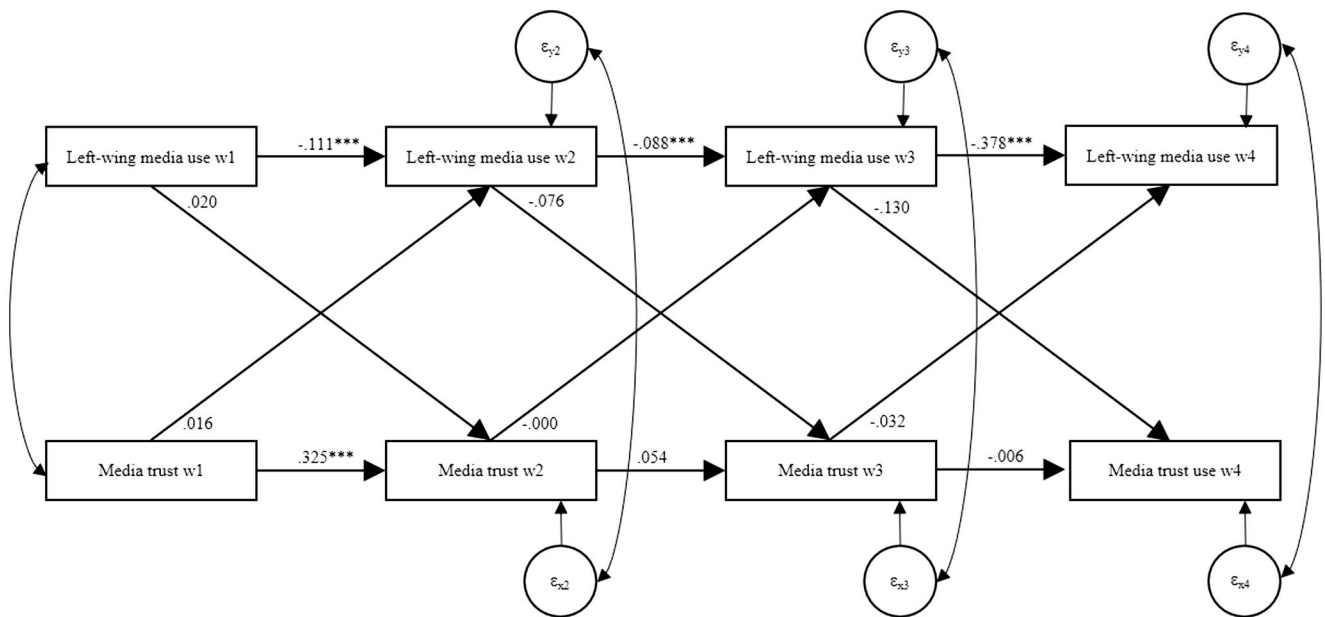


Figure 3. Reciprocal relationships between mainstream media trust and alternative left-wing media use.

Note. RI-CLPM with a linear slope for media trust, unstandardized B values, $\chi^2=108.118$; $df=8$, $p<.001$, CFI=.994; RMSEA=.059; * $p<.05$; ** $p<.01$; *** $p<.001$, $N=3,540$.

measure was targeted at the Swedish news media in general and not at TV or print outlets in particular.

Discussion and conclusions

This investigation aimed to decipher the longitudinal and causal mechanism(s) underlying the association between mainstream news media trust and mainstream and political alternative media use. As explained earlier, an effect of mainstream media trust on later exposure patterns is theoretically plausible given theories of selective exposure, while the

reverse effect of exposure on later trust can be accounted for using theories of social influence and persuasion. It is also possible that both mechanisms are at play simultaneously, leading to a reinforcing spiral.

Summarizing the findings, our study offers five key take-aways. First, the results corroborate that there is a relationship between media use and mainstream media trust. Second, the results suggest that this depends more on mainstream media trust influencing subsequent mainstream media use than the other way around. In other words, our findings support the notion that the key underlying theoretical mechanism

explaining the positive relationship between mainstream media use and media trust is selective exposure. Third, we also found some albeit inconsistent evidence that mainstream media use may influence subsequent media trust. This suggests that sometimes mainstream media are able to convince people of their trustworthiness and that people, over time, may become cultivated to understand trustworthy journalism in ways that are consistent with what mainstream media offer. Also, our results do not rule out that initial levels of mainstream media trust may be due to earlier socialization of mainstream media use on mainstream media trust. This is important, since it suggests that mainstream media can influence the degree to which people trust them. At least this holds at a general level, even if there might be groups that are harder to reach and persuade. This includes people with more populist, right-wing attitudes, who tend to perceive mainstream news media as biased and part of the elite (Fawzi, 2019; Noppari et al., 2019; Schwarzenegger, 2023).

The fourth take-away is related to the longitudinal dynamics, where the results suggest a growth trajectory for mainstream media trust, but not for mainstream news media use. This means that the effect of trust on use should be interpreted as a maintenance effect (Shehata et al., 2023), rather than a reinforcing spiral yielding clear upward or downward trends (Slater, 2007, 2015). The over-time growth trajectory for mainstream media trust suggests a pattern that is more consistent with the RSM. Past mainstream media use increases future trust in mainstream media (as explained above less consistently) in a manner that consequently makes trusting people more trusting through their mainstream media use. The theoretical explanation for this asymmetrical spiraling pattern and its theoretical implications will be discussed below.

The fifth take-away is related to the use of partisan alternative media, where findings show that the use of right-wing (but not left-wing) partisan alternative media use consistently leads to lower trust in mainstream media. It is noteworthy that the coefficient size gradually increases over time (from $b = -.078$ from wave 1 to wave 2 to $-.370$ from wave 3 to wave 4). This is consistent with findings from research showing that delegitimizing media criticism is quite prevalent in right-wing political alternative media (Figenschou & Ihlebaek, 2021; Roberts & Wahl-Jorgensen, 2020). The overall patterns for alternative right-wing media are more in line with the expectations of the RSM, with growth trajectories for both trust and use and negative (though not always significant) effects of mainstream media trust on alternative right-wing news use and of alternative right-wing media use on mainstream media trust. However, this pattern also does not resemble a perfect RSM spiral. One reason is because the coefficients are small and not consistently significant, while another is that the growth trajectory for right-wing media exposure shows decline accompanied by an overall increase in mainstream media trust over the four years of the study. The ensuing consequence on mainstream media trust should be a decrease on the aggregate level rather than the increase observed in our sample.

Altogether then, the findings suggest a complex longitudinal dynamic of an asymmetrical spiral in which trust increases gradually over time as a result of exposure, but trust-based selective exposure plays a more consistent effect in shaping future mainstream media use without increases in mainstream media use over time. There are multiple plausible

theoretical accounts for this pattern of findings. First, media selection decisions, though shaped by attitudes towards media, are also shaped by external factors unrelated to media trust. Examples include respondents' schedules and the preferences of their family members and many other factors (Webster, 2014). As Strömbäck et al. (2020, p. 145) maintain, "people use media, even news media and news genres, for other purposes than getting informed."

Second, and on the other hand, the stronger (especially in the case of the effect of right-wing news use on media trust) but more inconsistent effect of use on trust over time can be explained by what we know about media trust and how it is shaped by events and time. We know that media trust is probably shaped by fewer factors compared to exposure (Tsifti & Cohen, 2012) and our design accounts for all individual-level factors (such as education or ideology). However, we also know that at least in the short term, media trust may be influenced by the performance of the media (Major & Atwood, 1997) and by elite rhetoric about them (Van Duyn & Collier, 2019). The inconsistent effects of media use on mainstream media trust can perhaps be explained by the fact that media scandals and convincing attacks on media by politicians are not routine events in mainstream media. The lack of effect of mainstream media use on media trust between wave 1 and wave 2 and wave 2 and 3 may, for example, be due to the corona pandemic, since the media coverage in Sweden was criticized for, among other things, not being critical enough towards health authorities (Truedson, 2021). When harsh criticism occurs, it can be expected to exert some influence. In other words, an asymmetry in the reinforcing spiral in the case of media trust and mainstream media use can be explained by the different nature of media trust and media use.

Finally, beyond their importance for our understanding of the dynamics of media use and mainstream media trust, our findings carry theoretical implications for our understanding of reinforcing spirals more generally. While the possibility of asymmetry in reinforcing spirals has been acknowledged by Slater (2015, p. 383), very few studies have actually investigated and documented this possibility (e.g., Geusens & Beullens, 2021). As in the context of exposure to mainstream media, media exposure in other domains is shaped by a multiplicity of factors, which may lead to an asymmetry of the type we found in the case of media trust and mainstream news use. The other potential variables in an RSM feedback loop may at times be relatively immune to the effect of other factors, but at times influenced by more short-term and long-term factors compared to media use. Hence, our investigation joins others (Thomas et al., 2021) in establishing the utility of RI-CLP modeling in investigating and documenting complex reciprocal communication dynamics, in ways that significantly advance our understanding of questions of causality that has bothered media scholars for generations (e.g., Rossmann & Brosius, 2004).

In terms of asymmetry, the difference between the effects of using left- and right-wing media is also worth noting. Our interpretation is that this is because left-wing alternative media in Sweden is less "alternative" and more similar to mainstream news media than right-wing alternative media (Andersen et al., 2023; Holt, 2020; Shehata & Strömbäck, 2022; Theorin & Strömbäck, 2020). Along similar lines, research suggests that right-wing alternative media and political actors are typically much more hostile toward

mainstream news media than their left-wing counterparts (Holt, 2020; Figenschou & Ihlebaek, 2021; Nygaard, 2023). Thus, from a persuasion and media effects perspective, it makes sense that people who use right-wing but not left-wing alternative media subsequently trust mainstream news media less (Andersen et al., 2023).

While the investigation carries an important contribution to our understanding of the relationship between media use and mainstream media trust, it is not free of limitations. One has to do with the one-year time lag between panel waves. Determining the optimal time lag for the study of media selection and effects is difficult. As Thomas et al. (2021) argue, “for most situations and outcomes, we simply do not know how long it takes until media effects and selection appear.” A year may be a too large time interval for the effects under study. Similar to other non-experimental studies, it is also important to acknowledge that while our findings are consistent with causal claims, they do not offer decisive evidence supporting causality. However, our panel design helps us in establishing temporal order (Warner, 2017), and the RI-CLPM effectively controls for between-subject factors. Thereby, our findings offer “compelling evidence that is consistent with causality” (Warner, 2017, p. 121). Yet another limitation is that our mainstream media trust measure was targeted at the Swedish news media in general. While as explained above, our own data and previous studies (Tsfati et al., 2023) suggest that when people are asked about their trust in news media in general, they are thinking about traditional, mainstream news media, we cannot completely rule out that some respondents were not including alternative media when asked about their general news media trust.

Another obvious limitation is that we do not know to what degree we can generalize from our Swedish data to other contexts, although the association between media use and mainstream media trust has been observed in many different contexts. Sweden has a Democratic Corporatist media system (Brüggemann et al., 2014) and remains a country where traditional news media continues to hold a prominent place in citizens’ media diets (Newman et al., 2023). While alternative partisan media have become more important, exposure to such outlets is still relatively low in Sweden (Vliegenthart et al., 2024). This may suggest that some of the results are less likely to be replicated in other contexts. Still, we did observe enough variance in media trust and the findings for mainstream news media make theoretical sense in ways that lead us to expect that the findings for mainstream news media may replicate in other contexts. Obviously, these propositions should be put to an empirical test. As always, much work is left for future studies.

Supplementary material

Supplementary material is available at *Journal of Communication* online.

Data availability

The data underlying this article will be shared on reasonable request to the corresponding author.

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Conflicts of interest. None declared.

Notes

1. Some have argued that media trust is a multi-dimensional concept (Kohring & Matthes, 2007). However, the majority of empirical explorations view media trust as unidimensional, with some evidence even pointing out that a single-item trust in media measure performs almost identically to a multi-item scale (Tsfati et al., 2023, p. 5) consistent with the unidimensionality assumption.
2. Different scholars offer varying definitions of the concepts of mainstream and alternative media, with some scholars focusing on alternative production, distribution or funding, others on the distance between the values reflected and views expressed from elitist institutional power hubs in which mainstream news channels are embedded. According to more recent conceptualizations, the key dimension of the definition of alternative media is that they position themselves as an alternative to mainstream media with the intent of criticizing, correcting and complementing them (Holt et al., 2019). The focus of the current exploration is on partisan alternative media, a term referring to media outlets presenting the news with a right-wing or left-wing ideological slant (see Andersen et al., 2023). While we acknowledge that the concepts of mainstream-alternative/partisan media may be best understood as a continuum (Nygaard, 2023), most empirical studies simply separate mainstream and partisan alternative media, given that it is more parsimonious to do so.
3. While some readers may be concerned by the fact that some responses in Wave 1 and the entire Wave 2 were collected during the height of the COVID-19 pandemic, this actually presents an advantage to our design. This is because the pandemic added variance to the exposure and trust measures that are usually very stable (Scharkow, 2019), making this an excellent opportunity to detect longitudinal effects.
4. Despite the fact that these reliabilities are rather low, we created news exposure scales, given that the indicators in these cases are “causal indicators” rather than the usual “effects indicators”. In other words, when the latent construct is shaped by the indicators (that is, exposure to even one of the outlets constitutes overall mainstream exposure), we do not expect high consistency between the items, as acknowledged in the statistical literature.
5. While all three coefficients for the effects of media trust on mainstream news use were significant in the main analysis, the effect of Wave 3 trust on Wave 4 news was insignificant in the robustness check. On the other hand, while the effect of Wave 2 mainstream news use on Wave 3 media trust was not significant in the main analysis, it was significant in the robustness check. In the model for partisan right-wing media, while the effect of Wave 3 media trust of Wave 4 right-wing news use was significant in the main analysis, this effect was not significant in the robustness check. Instead, the effect of Wave 1 trust on Wave 2 partisan right-wing use was significant in the robustness check but not in the main analysis. Also, while the coefficient for the effect of Wave 3 trust on Wave 4 partisan right-wing news use was significant in the main analysis, it was not significant in the robustness test.

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